

AI, Human Cognition and Knowledge Collapse

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github.com/amchavez/ai-04-acemoglu

The paper and the agent's problem

Question

Can better personalized AI decisions today weaken the human effort that maintains collective knowledge tomorrow?

$$\max_{e \geq 0} \left\{ \begin{array}{l} f(0,0) + G(X)\Delta_G + G(X)G(Y)\Delta_X \\ - \frac{\varepsilon}{\varepsilon + 1} e^{(\varepsilon+1)/\varepsilon} \end{array} \right\}$$

$$Y = \sigma^{-2} + \lambda_I e + \tau_A$$

Two complementary inputs

- ▶ General knowledge: X about the common state
- ▶ Context-specific knowledge: Y about the current case

Effort and AI both raise private precision Y .

Externality. The agent captures the private return to effort, but not the general knowledge contributed to future cohorts.

Main result and conditions

$$U_{eX} = \lambda_I \Delta_X g(X) g(Y) > 0$$

$$U_{e\tau_A} = \lambda_I \Delta_X G(X) g'(Y) < 0$$

$X > 0, Y > 0, \Delta_X > 0, \lambda_I > 0, \varepsilon > 0$

General knowledge complements effort

Better common-state knowledge raises the value of learning about the particular case.

Agentic AI substitutes for effort

Both raise Y . Since $g'(Y) < 0$, AI lowers the marginal return to human learning.

Boundary note. At $X = 0$, $U_{e\tau_A} = 0$ and $g(0)$ is not finite. The strict cross-partials are interior statements.

What I did

1. Audited Observation 1

The economic mechanism is correct. Its strict derivative formulation requires $X > 0$. Discrete differences preserve the global interpretation weakly.

2. Relaxed $\Delta_I = 0$

Context-specific correctness now has value even when general knowledge is absent.

$$U_e = \lambda_I g(Y) [\Delta_I + G(X) \Delta_X] - e^{1/\varepsilon}$$

$$\Delta_I > 0 \Rightarrow e_{\Delta}(0, \tau_A) > 0 \\ \Rightarrow F_{\Delta}(0) > 0$$

Implication

Complete zero-knowledge collapse disappears, but low-knowledge steady states remain an open question.

Where I did not believe the AI

PENDING STUDENT PHOTO

`hand/observation1-boundary.jpg`

Cross-partials and the $X = 0$ boundary
check

Initial claim

Both cross-partials are strictly signed globally.

Hand check

At $X = 0$,

$$U_{e\tau_A} = 0, \quad g(0) \notin \mathbb{R}.$$

Verdict

Correct on the interior, incomplete at the boundary. The mechanism remains economically valid in a weak global sense.